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Matt Raymond

Generative ML + Nanochemistry

I use machine learning theory and pretrained models to solve grand challenges in computational nanochemistry

Education

Ph.D. in Signal and Image Processing and Machine Learning — Ann Arbor, MI
ECE Dept., University of Michigan *(GPA 3.92)* — 08/22-**04/27**

- Open source PyTorch/CUDA code for all papers available on GitLab: gitlab.eecs.umich.edu/mattrmd-public/
- Co-advised by Drs. Angela Violi (MechE) and Clayton Scott (ECE)
- Thesis topic: **Generative Nanoparticle Design**

M.Sc. in Computer Science — Ann Arbor, MI
CSE Dept., University of Michigan *(GPA 3.91)* — 08/20–04/22

B.Sc. in Computer Science — Orange, CA
Chapman University *(GPA 3.86, magna cum laude)* — 08/16–05/20

Research Experience

Directed Study & Research — Ann Arbor, MI
Drs. Angela Violi (MechE) and Clayton Scott (ECE) — 01/21–present

- Develop algorithm to sample phase-separating proteins from pretrained diffusion models without retraining
- >10× accelerated screening/simulation throughput during collaborations with chemists and plasma physicists
- Perform 20× sparser selection of chemical features with a novel, multi-task approach using boosted trees
- Supervise undergrad/masters research in geometric deep learning, deep Gaussian processes, and active learning
- Co-author grant proposals, from small departmental grants to multi-million dollar research centers

Directed Study & Research — Ann Arbor, MI
Dr. Mania Meibody (Architecture & Urban Planning) — 07/20–11/21

- Used Babylon.js, Node.js, React, and Redis to develop a metaverse-like pedagogical tool for architecture

Directed Study & Research — Orange, CA
Drs. Eric Linstead and Justin Walsh — 01/20–05/20

- Used facial recognition to map the interpersonal dynamics of astronauts for the ISS Archeology project

Work Experience

Data Science Intern — Troy, MI
Toyoda Gosei North America — 05/20

- Hired, but the internship was canceled because of COVID-19

Instrument Programmer — Long Beach, CA
Lotus Instruments — 09/19–11/19

- Implemented controls and data transfer protocols for government-contracted gas chromatography instruments

Software Engineering Intern — Troy, MI
Toyoda Gosei North America — 05/19–08/19

- Saved 2k man-hours and $60k per year through automated purchase order tracking with ASP.NET
- Collaborated with Cost Management to solidify requirements and return on investment

Technical Skills

PyTorch, CUDA, Slurm, Git, Docker, Python, Jax/Flax, LaTeX, WandB, diffusion models, geometric deep learning, interdisciplinary collaboration, transformers, optimization, linear algebra, probability theory, equivariant neural networks, active learning, probabilistic modeling

Journal Articles

M. Raymond, J. C. Saldinger, P. Elvati, and A. Violi, "Universal feature selection for simultaneous interpretability of multitask datasets," *Journal of Cheminformatics*, vol. 18, no. 1, p. 23, 2026. 10.1186/s13321-025-01096-z, https://gitlab.eecs.umich.edu/mattrmd-public/gendesc/gendesc-main-repo

M. Raymond, P. Elvati, J. C. Saldinger, J. Lin, X. Shi, and A. Violi, "Machine learning models for Si nanoparticle growth in nonthermal plasma," *Plasma Sources Science and Technology*, vol. 34, no. 3, p. 35014, 2025. 10.1088/1361-6595/adbae1, https://gitlab.eecs.umich.edu/mattrmd-public/ntp-silicon

J. C. Saldinger, **M. Raymond**, P. Elvati, and A. Violi, "Domain-agnostic predictions of nanoscale interactions in proteins and nanoparticles," *Nature Computational Science*, vol. 3, no. 5, pp. 393–402, 2023. 10.1038/s43588-023-00438-x, https://gitlab.eecs.umich.edu/violigroup/ml/neclas/-/releases

Conference Proceedings

M. Raymond, P. Elvati, and A. Violi, "Mapping Kinetic Bottlenecks in Soot Precursor Growth via Graph Analysis of Stochastic Trajectories," in *41st International Symposium on Combustion*, 2026. Accepted

Workshops

M. Raymond, Y. Zhu, J. Zhang, A. Violi, and C. Scott, "Joint Diffusion Sampling via Positive-Unlabeled Guidance for Multi-Modal Data," in *ICML 2025 Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences*, 2025. URL: https://openreview.net/forum?id=1o89PN84NH

M. Raymond, A. Violi, and C. Scott, "Joint Optimization of Piecewise Linear Ensembles," in *IEEE International Workshop on Machine Learning for Signal Processing*, 2024. 10.1109/MLSP58920.2024.10734791, https://gitlab.eecs.umich.edu/mattrmd-public/joplen-repositories/joplen/

Invited Talks

Joint Diffusion Sampling via Positive-Unlabeled Guidance for Multimodal Data — Ann Arbor, MI
University of Michigan, Michigan Student Symposium for Interdisciplinary Statistical Sciences — 03/26

From Proteins to Nanoparticles: Domain-Agnostic ML for the Nanoscale — Ann Arbor, MI
University of Michigan, Tools and Technology Seminar Series — 03/25

Joined Diffusion Models for Nanoparticle Generation — Orange, CA
Chapman University, Fowler School of Engineering Seminar Series — 11/24

Machine Learning Models for Nanoparticle Growth in Nonthermal Plasmas — Minneapolis, MN
University of Minnesota, Dusty Plasma Workshop — 05/24

Poster Presentations

Joint Diffusion Sampling via Positive-Unlabeled Guidance for Multi-Modal Data — Vancouver, CA
ICML, 2025 Workshop on Foundation Models for Life Science — 07/25

Joined Diffusion Models for Nanoparticle Design — Chicago, IL
University of Chicago, Midwest Machine Learning Symposium — 06/25

Joined Diffusion Models for Nanoparticle Design — Ann Arbor, MI
University of Michigan, 2nd Conference on Foundation Models and AI Agents for Science — 05/25

Accel. Drug Discovery: Modeling Nanoscale Interactions in Protein-Nanoparticle Systems — Ann Arbor, MI
University of Michigan, DATA Spring 2025 Industy Advisory Board meeting — 03/25

Machine Learning for Knowledge Transfer in Nanomedicine — Ann Arbor, MI
University of Michigan, e-Health and Artificial Intelligence — 09/24

Joint Optimization of Piecewise Linear Ensembles — London, UK
Imperial College London, IEEE Machine Learning for Signal Processing Workshop — 09/24

Joint Optimization Significantly Improves Gradient Boosting — Minneapolis, MN
University of Minnesota, Midwest Machine Learning Symposium — 05/24

★ **Best Poster Award** Press Release Official Link

ML Boosts Efficiency in Atomic Nonthermal Plasma Simulations — Minneapolis, MN
University of Minnesota, Dusty Plasma Workshop — 05/24

Joint Optimization of Linear Ensembles — Ann Arbor, MI
University of Michigan, Michigan Student Symposium for Interdisciplinary Statistical Sciences — 03/24

⋆ **Best Poster Nominee**

A Taste of Your Own Medicine: Tracing Butyrate Production In The Gut — Ann Arbor, MI
University of Michigan, e-Heath and AI Student Symposium — 09/23

Preventative Healthcare via Accessible Data Tools — Orange, CA
Chapman University, Student Scholar Symposium — 05/20

Awards and Honors

Top 15 Poster Presentation — Minneapolis, MN
University of Minnesota, Midwest Machine Learning Symposium — 04/24

- Official Link Press Release

Excellence in ECE Honor Roll — Ann Arbor, MI
University of Michigan — 05/24

- For managing the SPEECS graduate student seminar during the 2023-2024 academic year: Press Release

Music Minor Award — Orange, CA
Chapman University — 05/20

- In recognition of outstanding contributions to the conservatory (*i.e.*, tutoring)

Provost List, 8 semesters — Orange, CA
Chapman University — 08/16–05/20

- $\geq$3.800 term GPA while enrolled in ≥12 graded credits: Source

Grants and Funding

J. Robert Beyster Computational Innovation Graduate Fellowship — Ann Arbor, MI
University of Michigan — 07/24

- Tuition, stipend, and health insurance for a year
- "Cutting-edge research linking high-performance computing to applications of societal importance." Press

e-HAIL Dataset Creation Funding ($5,000 grant) — Ann Arbor, MI
University of Michigan — 11/25

e-HAIL Summer Student Support Funding ($12,000 stipend) — Ann Arbor, MI
University of Michigan — 01/24

e-HAIL Summer Student Support Funding ($12,000 stipend) — Ann Arbor, MI
University of Michigan — 01/23

Chancellor's Scholarship, 8 semesters — Orange, CA
Chapman University — 08/16–05/20

Reviewing

Information Fusion (IF 15.5) — 02/26

NeurIPS 2025 Workshop on Foundation Models for Life Science — 09/25

NeurIPS 2025 Workshop on AI for Science — 09/25

ICML 2025 Workshop on Foundation Models for Life Science — 06/25

Teaching and Mentorship

Graduate Student Instructor — Ann Arbor, MI
University of Michigan — 08/25–Present

- Hold office hours and supplemental lectures each week
- Collaboratively write exam problems and teaching materials
- Contribute math animations (*a lá* 3Blue1Brown) for a new course under development

- Principles of Mach. Learning (EECS 453), Prob. Methods in Engineering (EECS 301)

Programming Instructor — Irvine, CA
Coding Minds Academy — 07/20–11/20

STEM Tutor — Orange, CA
Chapman University — 09/19–05/20

- Group tutoring for Calculus 1-3, Discrete Math, Visual Programming, Assembly Language, Data Structures and Algorithms, Programming Languages, Genetic Biology, and Molecular Genetics

Supplemental Instructor for Data Structures and Algorithms — Orange, CA
Chapman University — 09/19–12/19

- Held office hours and supplemental lectures each week

Music tutor — Orange, CA
Chapman University — 01/17–05/20

- Private tutoring for Music Theory, Music History, Aural Skills, and Music Technology
- Led group review sessions for music history

Service

Proofreader for a Linear Algebra Textbook — Ann Arbor, MI
Cambridge University Press — 05/23–09/23

- Proofread a linear algebra textbook from Cambridge University Press for Drs. Jeffery Fessler and Raj Rao
- Independently verified proofs, suggested improvements for clarity, and caught LaTeX errors

Coordinator, Signal Processing in EECS (SPEECS) seminar — Ann Arbor, MI
University of Michigan, SPEECS — 08/23–05/24

Coordinator, Machine learning theory reading group — Ann Arbor, MI
University of Michigan, ML Theory Reading Group — 01/22–02/22

Planning Committee Member, Engineering Research Symposium (ERS) — Ann Arbor, MI
University of Michigan, ERS — 07/20–02/21

Coordinator, COVID-19 Study Group — Virtual
Chapman University — 05/21–07/21

- Coordinated an MIT OpenCourseWare study group for students who lost summer internships due to COVID-19
- Operating Systems , Microeconomic Theory and Public Policy , and Probability Theory

Vice-President, FPV Drone Club — Orange, CA
Chapman University — 01/20–05/20

Class Projects

The Implicit Bias of Gradient Descent on Separable Multiclass Data — Ann Arbor, MI
U-M Course: EECS 598, 559 — 12/22, 05/23

- Developed a proof sketch for the implicit bias of gradient descent on separable data with multiclass PERM losses
- Showed numerically that our conjecture holds for certain well-known loss functions

Real-Time Distributed Learning in Connected & Autonomous Vehicles (CAVs) — Ann Arbor, MI
U-M Course: EECS 571 — 12/21

- Designed distributed learning protocol for sparse gradient propagation
- Implemented simulated learning environment in Tensorflow
- Demonstrated superior generalization, with fewer assumptions than Federated Learning

Domain Exploration Through Artificial Curiosity — Ann Arbor, MI
U-M Course: EECS 545 — 12/20

- Developed simulated Martian terrain for training and evaluation
- Implemented Shmidhuber's definition of artificial curiosity using convolutional auto-encoders
- Performed evaluation against naive models to illustrate effectiveness of artificial curiosity

Needlecast: On-the-Fly Reconfiguration of Spacecraft Flight Software — Ann Arbor, MI
U-M Course: EECS 587 — 12/20

- Collaborated with NASA staff to draft specifications for protocols

- Designed a library for booting NASA core Flight System (cFS) applications on-the-fly
- Implemented Needlecast (see here) as a plug-and-play header file for NASA cFE
- Developed a simulated network switch and web interface for straightforward debugging

Preventative Healthcare via Accessible Data Tools Orange, CA
Chapman Course: CPSC 353 05/20

- Developed a web application to rank establishments by the number of COVID-19 cases in their area
- Utilized REST APIs and front-end web development to develop a user-friendly interface

AI-Driven Contemporary Archaeology for the International Space Station Orange, CA
Chapman Course: CPSC 393 01/20

- Analyzed project requirements with Dr. Justin Walsh (co-PI of ISS Archeology)
- Used facial recognition to map the interpersonal dynamics of astronauts